

Exam Date & Time: 03-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech V Semester - Make up Examination, Jan-2025
Fundamentals of Machine Learning [CSE-3173]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Classify the different types of Machine Learning and provide examples for each category. Also briefly describe a scenario where each type might be applied (3)
 - A)
 - B) A machine learning team is testing a spam filter and finds that an email has a 30% chance of being spam. The team also determines that if an email is spam, there is a 70% chance it will contain the word "offer." However, if an email is not spam, there is only a 10% chance it will contain "offer." Given that an email contains "offer," what is the probability that it is spam? (5)
 - C) Given the following covariance matrix of a 2D dataset:
$$\Sigma = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$$
calculate the eigenvalues of Σ . (2)
- 2) List key features of the K Nearest Neighbor algorithm. (2)
 - A)
 - B) Compare logistic regression with regularization and without regularization. List two types of regularization and their impact on model performance. (4)
 - C) (4)

Consider the following dataset with two features X_1 and X_2 (both continuous) and a binary target, Y .

Sample	X_1 (Time of the Day)	X_2 (Traffic Condition)	Y (Commute Fast)
S_1	Morning	High	No
S_2	Morning	High	No
S_3	Afternoon	Medium	Yes
S_4	Evening	High	No
S_5	Evening	Low	Yes
S_6	Morning	Low	Yes
S_7	Afternoon	Medium	Yes
S_8	Afternoon	High	No
S_9	Evening	Medium	Yes
S_{10}	Morning	Medium	Yes
S_{11}	Evening	Low	Yes
S_{12}	Afternoon	Low	Yes
S_{13}	Morning	Medium	Yes
S_{14}	Evening	Medium	Yes

Using the **CART algorithm** and **Gini Impurity** as the criterion, create the decision tree.

- 3) Consider an artificial neural network with one input layer, one hidden layer, and one output layer. The input layer has two neurons, the hidden layer has three neurons, and the output layer has one neuron. The weights from the input layer to the hidden layer are:
- A)

$$\text{Weights}_{\text{input to hidden}} = \begin{bmatrix} 0.5 & -0.4 \\ 0.3 & 0.1 \\ -0.6 & 0.2 \end{bmatrix}$$

The weights from the hidden layer to the output layer are:

(3)

$$\text{Weights}_{\text{hidden to output}} = \begin{bmatrix} 0.4 \\ -0.3 \\ 0.2 \end{bmatrix}$$

The input values are $X=[2, -1]$, and the sigmoid activation function is applied in the hidden layer. The output layer uses a simple linear activation function (output = input). Calculate the output of the network.

B)

(4)

Consider a dataset with two features, X_1 and X_2 , and a binary target variable Y . We have the following training data:

Sample	X_1 (Type of Day)	X_2 (Temperature)	Y (Go for a Run)
S1	Weekday	High	Yes
S2	Weekday	Low	No
S3	Weekend	High	Yes
S4	Weekend	Low	Yes
S5	Weekday	Low	No
S6	Weekday	High	Yes
S7	Weekend	Low	Yes
S8	Weekend	High	Yes
S9	Weekday	Low	No
S10	Weekday	High	No
S11	Weekend	Low	Yes
S12	Weekday	High	Yes
S13	Weekend	Low	Yes
S14	Weekday	High	No

You are given a new instance with $X_1 = \text{Weekend}$ and $X_2 = \text{High}$. Using Naive Bayes, calculate the probability that $Y = \text{Yes}$ or $Y = \text{No}$, and classify the new instance.

- C) In a hospital, there is a Bayesian Belief Network used to monitor the probability of emergency systems working correctly during a power outage. The network consists of three events: (3)

1. A : Underwater Pressure Drop Detected
2. B : Acoustic Sensor Triggered
3. C : Leak Alarm Activated

The probability values are given as follows:

- The probability that a pressure drop is detected, $P(A=\text{True})=0.03$, and $P(A=\text{False})=0.97$.
- If a pressure drop is detected, the probability that the acoustic sensor will also be triggered, $P(B=\text{True} \mid A=\text{True})=0.8$, and there's no pressure drop, the probability of the acoustic sensor being triggered is, $P(B=\text{True} \mid A=\text{False})=0.1$.

The probability that the leak alarm activates depends on both the pressure drop and acoustic sensor states:

- $P(C=\text{True} \mid A=\text{True}, B=\text{True})=0.95$
- $P(C=\text{True} \mid A=\text{True}, B=\text{False})=0.4$
- $P(C=\text{True} \mid A=\text{False}, B=\text{True})=0.2$
- $P(C=\text{True} \mid A=\text{False}, B=\text{False})=0.01$

Calculate the probability that the leak alarm is activated, $P(C=\text{True})$, regardless of the pressure drop or acoustic sensor status. Give the Bayesian Belief Network with corresponding tables for each node.

- 4) Given a dataset with high dimensionality, you are tasked with building a machine learning model. Discuss how the bias-variance trade off would influence your choice of model complexity. Apply this concept to suggest appropriate strategies for improving model performance in both underfitting and overfitting scenarios (3)
- A)
- B) List three different ensemble learning techniques. Using process flow and diagrams illustrate how each of these methods enhances the performance of machine learning models. (5)
- C) List the steps of Random Forest algorithm. Using suitable dataset illustrate how Random Forest algorithm can be used for building and predicting a model. (2)

- 5) You are working with a dataset consisting of two features, x_1 and x_2 . The following table represents the data points:

- A)
- a) Reduce the dimension from 2 to 1 using the Principal Component Analysis (PCA) algorithm.
- b) Show the geometrical representation of the dataset before and after applying Principal Component Analysis (PCA). (5)

Data Point	x_1	x_2
D1	4	11
D2	8	4
D3	13	5
D4	7	14

- B) Consider a dataset with the following three data points in a 2D space:

Sample	X_1	X_2
P_1	1	2
P_2	2	1
P_3	3	4
P_4	5	7
P_5	3	3
P_6	6	5

(5)

Create $k=2$ clusters from these points using the K-Means algorithm, then show all the steps and computations that went into creating the final clusters.

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